

Answer Key

Quiz 2

Weeks 3–4: How LLMs Work, and Principles of Effective Prompting

1. **B** Tokens are not words. Common words are often one token; rarer words fragment; spaces are usually attached to the following word.
2. **B** A fixed vocabulary of whole words fails on any unseen word. Sub-word pieces mean the system is never stuck — new names, technical terms and typos can all be assembled.
3. **B** Agglutinative stacking produces long words that appeared rarely in training, so they fragment. Consequences: the context fills faster, metered usage costs more, and coverage is thinner.
4. **B** Letter counting asks about something the system never perceived. This is a predictable consequence of tokenisation, not a general weakness at arithmetic.
5. **B** A probability distribution over the whole vocabulary, from which one token is selected — then the process repeats. Nothing is retrieved and no rules are applied.
6. **B** Always selecting the most probable token produces flat, repetitive text, so randomness is introduced deliberately. Often exposed as “temperature”.
7. **B** Consistency measures how strongly a pattern is represented, not whether it is true. A widely repeated error is reproduced very consistently.
8. **C** The system optimises plausibility. Plausible usually correlates with true, but nothing checks, so where they diverge nothing notices. This is why it cannot simply be patched out.
9. **C** Instructions change tone and shape, never available information. Asking for authority produces authority — which makes a wrong answer more convincing, not less.
10. **C** Wrong shape means Format. Wrong content means Context or Task. Correct but lifeless means Examples. Format is the cheapest lever and the most neglected.